
HOMEWORK DAY 7– *Combine Function (Section 6.4)*

1. Answer the following for the given functions and find their domains.

$$r(x) = -3x \quad p(x) = x^2 + 3x \quad q(x) = \sqrt{1-x}$$

(a) $(r - p)(x)$

(b) $\left(\frac{q}{p}\right)(x)$

(c) $\left(\frac{r}{q}\right)(x)$

2. Answer the following for the given functions and find their domains.

$$s(x) = \frac{x-2}{x^2-9}, t(x) = \frac{x-3}{2-x}$$

(a) $(s \cdot t)(x)$

(b) $\left(\frac{s}{t}\right)(x)$

3. Use **Ex10** functions to find $(g - f)(x)$ and its domain $f(x) = \frac{1}{x-1}$, $g(x) = \frac{x-2}{x^2-1}$

4. Answer the following for the given functions and find their domains.

$$q(x) = \frac{1}{x-10}, \quad p(x) = x^2 - 9x, \quad n(x) = x - 5, \quad m(x) = \sqrt{x+8}$$

(a) $(m \circ n)(x)$

(b) $(q \circ n)(x)$

(c) $(q \circ p)(x)$

(d) $(q \circ m)(x)$

5. Find $(f \circ g)(x)$ and its domain $f(x) = \frac{9}{x-1}$, $g(x) = \frac{9}{x^2}$

6. Given functions $f(x) = \sqrt{x+1}$, $g(x) = \frac{9}{x^2}$ Answer the following.

(a) $(f \circ f)(3)$

(b) $(f \circ g)(2)$

(c) $(g \circ g)(2)$

(d) $(g \circ f)(x)$ and its domain

HOMEWORK DAY 8 – *Inverse Function (Chapter 9 Section 9.1)*

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